

面向视频游戏行为可信度的可自动化评估方法

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摘要

经典的可信智能体评估方法耗时较长, 因为它们需要多位人类评判者来评估智能体。这些方法非常适合验证关于新可信行为模型的研究成果。然而, 在实现过程中, 大量实验有助于提升智能体特征的可信度。我们提出了一种方法, 旨在评估智能体的行为在多大程度上与人类行为相似。通过用向量表示行为, 我们可以存储为人类计算的数据, 然后评估任意数量的智能体, 无需再依赖人类参与。我们进行了一项测试实验, 结果表明, 即便按照我们的方法进行简单的评估, 也能揭示出相当可信的智能体与人类之间的差异。尽管如实验所示, 结果分析可能较为困难, 但该方法前景可观。

引言

绝大多数视频游戏都包含计算机控制的虚拟角色, 也称为智能体, 以丰富其环境。为了使游戏能够维持玩家的沉浸感 (Bates 1992), 这些智能体应当具备可信度。然而, 这个术语有两种不同的含义: 其一, 是扮演好角色的能力; 其二, 是与人类玩家难以区分的能力 (Livingstone 2006)。我们的目标是让智能体扮演玩家的角色, 因此在本文中采用后一种定义。

在建模或实现智能体的行为时, 评估是一个重要步骤。通过调整模型参数或修改模型本身, 对于改进最终结果非常有帮助。

评估对于验证工作几乎是强制性的, 并且应当能够回答: 智能体在多大程度上达到了目标? 或者它是否比之前的工作更好? 然而, 可信智能体的评估相当困难, 因为可信度是基于观察者的感受。许多研究处理了这个问题, 例如 (Mac Namee 2004; Livingstone 2006; Gorman et al. 2006)。然而, 这些研究仍然依赖人类来评判智能体。因此, 这种方法无法用于评估大量不同的行为。此外, 使用问卷调查进行评估也受到了批评 (Slater 2004)。

了解了这些问题后, 我们提出了一种采用不同方法: 我们的目标是尽可能减少人类干预, 同时仍能帮助研究人员评估人工行为的可信度。其目标是拥有一种可以自动化的方法, 从而能够用于优化算法或夜间测试。为此, 我们选择衡量人工行为与人类行为的相似性, 这是一个接近可信度的概念。当然, 这种方法并非旨在取代经典方法, 而是提供一种补充途径, 使最终结果看起来更可信。

评估方法提议

原理

评估的总体原则是计算体现人类与智能体特征行为特征的向量。我们称这些向量为“签名”, 因为它们应被设计用于表征某种行为。通过测量智能体签名与人类签名之间的距离, 应当能够判断其行为是否类似于人类行为。

AUTOMATABLE EVALUATION METHOD
ORIENTED TOWARD BEHAVIOUR BELIEVABILITY
FOR VIDEO GAMES

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KEYWORDS

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ABSTRACT

Classic evaluation methods of believable agents are time-consuming because they involve many human to judge agents. They are well suited to validate work on new believable behaviours models. However, during the implementation, numerous experiments can help to improve agents’ believability. We propose a method which aim at assessing how much an agent’s behaviour looks like humans’ behaviours. By representing behaviours with vectors, we can store data computed for humans and then evaluate as many agents as needed without further need of humans. We present a test experiment which shows that even a simple evaluation following our method can reveal differences between quite believable agents and humans. This method seems promising although, as shown in our experiment, results’ analysis can be difficult.

INTRODUCTION

The vast majority of video games features computer controlled virtual characters, also known as agents, to enrich their environment. For games to be able to suspend disbelief (Bates 1992), those agents should be *believable*. However, this term have two distinct meanings: first, the ability to be a good role-player and second, the ability to be indistinguishable from a human player (Livingstone 2006). We aim at agents taking on the role of players, so we will use the latter definition in this paper.

When modelling or implementing an agent’s behaviour, an important step is evaluation. It can be very useful to improve the final result by tuning the

model’s parameters or by modifying the model itself. Evaluation is almost mandatory to validate work and should be able to answer: to what extent does the agent meet the objectives? or is it better than what have been done before? However, evaluation of believable agents is rather difficult because believability is based on the observers’ feelings. Many studies deal with this problem like (Mac Namee 2004; Livingstone 2006; Gorman et al. 2006). However, they still rely on humans to judges agents. As a result, this kind of method cannot be used to evaluate a large number of different behaviours. Moreover, the use of questionnaires for evaluation is criticised (Slater 2004).

Knowing these problems, we propose a method using a different approach: we aim at reducing human intervention as much as possible still helping researchers assessing artificial behaviours’ believability. The objective is to have a method which could be automatised and thus which can be used in optimisation algorithms or for tests during the night. For this, we chose to measure artificial behaviours’ likeness to human behaviours, a concept close to believability. Of course, this method does not aim at replacing classic ones but rather offers a complementary approach so that the final result looks more believable.

EVALUATION METHOD PROPOSITION

Principle

The evaluation’s general principle is to compute vectors which are features of humans’ and agents’ behaviours. We call those vectors “signatures” as they should be designed to be representative of a behaviour. By measuring the distance between an agent’s signature and humans’, it should be possible to tell if its behaviour looks like a human behaviour or not.

协议

以下是评估智能体相对于人类行为的协议主要阶段：

1. 定义行为特征及它们之间的指标；
2. 监测人类（越多越好）并计算特征；
3. 监测待评估的智能体，并在与监测人类相同的条件下计算特征；
4. 计算每个智能体特征与其人类对应特征之间的距离。

第一部分：人类行为特征

第一部分极具挑战性，因为我们需要找到对人类而言方差较低、同时仍能检测出不可信行为的特征。当人类被赋予极大自由度时，这可能会成为问题，因为他们的行为可能差异很大。为了确定良好的特征，可能需要用少量人类多次运行协议，并修改特征和/或指标，使其具备良好的属性。

人类行为特征与人类行为的距离

对于第二步和第三步，人类和智能体必须在相同条件下进行研究。由于智能体不与人类对抗，如果虚拟角色共存于同一环境中，我们需要做出让步：在第二步中，人类聚集在一起；在第三步中，智能体聚集在一起。如果在第二步中引入智能体，特征可能会因反映人类与不可信智能体之间的交互而产生偏差。

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人类行为特征与人类行为的距离

我们提出的方法是可自动化的。实际上，步骤3和步骤4可以在无需进一步人工干预的情况下重新执行。然而，自动化并不意味着快速，因为步骤3可能需要一些时间。为了确定实验的持续时间，需要进行一项研究来测试特征需要多长时间才能稳定下来。

监控智能体与人类

如协议所述，我们必须同时监控人类和智能体，以便计算签名。为了完全独立于内部架构，我们选择采用与评判者从受试者身后观察相同的视角。因此，在众所周知的感知-决策-行动循环中，我们只能访问动作和感知。我们假设该循环同样可对人类进行观察，从而能以相同方式比较智能体特征与人类的感知及动作。

监控的原理是在模拟过程中跟踪这些感知和动作的子集，以构建签名。例如，在第一人称射击游戏中，我们可以追踪基本动作和感知，比如角色是否跳跃以及是否看到敌人。这样就有可能构建一个非常简单的签名，用于衡量在敌人可见时的跳跃比例。

人类行为特征与人类行为的距离

主要有三类特征：基于感知的特征、基于动作的特征，以及链接动作与感知的特征。基于感知的特征用处不大，因为玩家是根据智能体的动作来评判它们的。基于动作的特征可能有用，但可能过于简单，无法解释复杂的行为。最后，链接动作与感知的特征是最有趣的，因为它们可能在决策中发现模式。然而，存在一个难点：如果在感知中监控到一条信息，并不能确定智能体或人类是否注意到了这条信息。

实验

在实验中，我们使用了《虚幻竞技场2004》这款游戏。它是一款第一人称射击游戏，换句话说，每个玩家或智能体控制着一个独特的虚拟角色，并通过其视角观察世界。该角色可以进行（但不限于）拾取物品（武器等）、移动（行走、跳跃等）和使用武器射击。每个角色都有一定数量的生命值，也称为生命点数：每次角色被敌人火力击中时，当前生命值会减去一定数额。当生命值降至零时，角色会“死亡”，但通常可以在虚拟环境中的另一个位置重新出现。尽管这个概念看似非常简单，但对于智能体来说，模仿人类行为可能颇具挑战性。

人类行为特征与人类行为的距离

本实验的主要目的是提供一个具体示例，并展示该方法的实用性。因此，这些签名设计得较为简单，尚不足以用于真实评估。此外，为简化起见，本示例实验仅尝试评估虚幻竞技场2004中智能体运动行为的可信度。以下是签名内容：

- 速度变化角度：一个20维向量，其 i 分量的值为测量到的角度约在 i 至 $\vec{V}_t$ 与 $\vec{V}_{t+1}$ 之间的次数。请注意，该向量仅有20个分量，因此角度值20对应整圈旋转，10对应半圈旋转，以此类推。该向量随后会被归一化处理。
- 相对于方向的速度：20维向量，第 i 个分量的值是在 $\vec{V}_t$ 到 $\vec{D}_t$ 之间近似角度 i 的测量次数。与之前

Protocol

Here are the main stages of the protocol to evaluate an agent with respect to humans:

1. Define behaviour signatures and metrics between them;
2. Monitor humans (the more the better) and compute the signatures;
3. Monitor the agent to be evaluated and compute the signatures in the same conditions as the monitored humans;
4. Compute the distance between each agent’s signature and its human equivalent.

人类行为特征与人类行为的距离

The first part is very challenging because we need to find signatures having a low variance for humans, still being able to detect non-believable behaviours. This could be a problem when humans are given a great freedom because their behaviour can vary widely. To determine good signatures, it can be necessary to run the protocol several times with few humans and to modify signatures and/or metrics so that they have nice properties.

人类行为特征与人类行为的距离

For the second and the third steps, human and agents must be studied under the same conditions. Since agents do not play against humans, if virtual characters coexist in the same environment we have to make a concession: for the step 2, humans are together and for the step 3, agents are together. If we introduce agents in step 2, signatures could be flawed by reflecting interaction between humans and non-believable agents.

人类行为特征与人类行为的距离

We presented our method as been automatable. Indeed, step 3 and 4 can be redone without further need of humans. However, automatable does not mean fast as step 3 can take some time. To determine the experiment’s duration, a study should be done in order to test how much time the signatures need to stabilise.

Monitoring agents and humans

As explained in the protocol, we must monitor both humans and agents so that we can compute signatures. To be totally independent from the internal architecture, we chose to take the same point of view as a judge looking over the subject’s shoulder. Therefore, in the well-known perception-decision-action loop we can only have access to the actions and the perceptions. We assume that the loop can be observed for human as well so that we can compare agents’ and humans’ perceptions and actions in the same manner.

The principle of the monitoring is to keep track of a subset of these perceptions and actions during simulations so as to build the signatures. For example, in a first person shooter, we could track basic actions and perceptions such as if the actor is jumping and if it sees an enemy. It could then be possible to build a very simple signature measuring the proportion of jumps when an enemy is visible.

人类行为特征与人类行为的距离

There are three main kind of signatures: perception-based, action-based and those linking actions to perceptions. Perception-based signatures are not very useful because players judges agents on their actions. Action-based can be useful but may be too simple to explain complex behaviours. Finally, signatures linking actions to perceptions are the most interesting ones because they may find patterns in the decisions. However, there is a difficulty: if an information is monitored in perceptions, it is not sure that the agent or the human noticed this information.

EXPERIMENT

人类行为特征与人类行为的距离

For the experiment, we used the game Unreal Tournament 2004. It is a first person shooter game, in other words each player or agent controls a unique virtual character and sees through its eyes. The character can, non-thoroughly , grab items (weapons, ...), move (walk, jump, ...) and shoot with a weapon. Each character have an amount of hit points, also known as life points: each time an actor is hit by an enemy fire, a certain amount of hit points are subtracted to the current value. When hit points reaches zero, the character “dies” but can usually reappear at another place in the virtual environment. Although the concept can seem very basic, it can prove challenging for agents to mimic humans.

人类行为特征与人类行为的距离

This experiment’s main objective is to give a concrete example and to show the interest of the method. Therefore, the signatures are simple and would be insufficient for a real evaluation. Moreover, for the sake of simplicity, this sample experiment try to assess only the believability of Unreal Tournament 2004 agents’ movements. Here are the signatures:

- Velocity change angle: 20-dimensions vector, the value of the i -th component is the number of time we measured an angle of approximatively i between $\vec{V}_t$ and $\vec{V}_{t+1}$. Note that the vector as only 20 components so an angle of 20 corresponds to a whole turn, 10, half turn, etc. This vector is then normalised.
- Velocity relative to the direction: 20-dimensions vector, the value of the i -th component is the number of time we measured an angle of approximatively i between $\vec{V}_t$ and $\vec{D}_t$. The same unit as the

签名相同的角度单位被使用。该向量随后进行归一化。

其中 t 是当前时间步长， $t + 1$ 是下一个时间步长，时间步长每125毫秒出现一次。 $\vec{V}$ 是移除了 z 分量的速度向量， $\vec{D}$ 是指向角色瞄准点（不含 z 分量）的向量（图1）。在签名中忽略 z 分量，是因为这样可以在不损失过多信息的情况下简化签名。

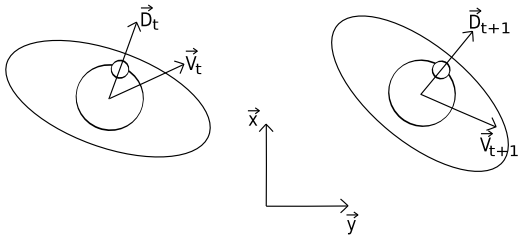


图1：在两个连续时间步长 t 和 $t + 1$ 下，从俯视图观察的虚拟角色示意图。 $\vec{D}$ 是指向角色瞄准点的方向向量， $\vec{V}$ 是速度向量。

实验包含两个步骤：

1. 6名低/中级人类玩家进行了一场20分钟的死亡竞赛游戏，期间计算了签名。我们选择20分钟的时长，是因为签名往往在15至20分钟后趋于稳定。死亡竞赛遵循基本规则，目标是尽可能多地“击杀”其他玩家。每名人类玩家使用键盘和鼠标控制虚拟角色，并通过经典台式电脑屏幕观察虚拟环境。

2. 为8个具有8种不同技能水平的智能体计算签名。这些智能体在相同地形中，遵循相同规则并在完全相同的时长内，以6智能体对战的形式相互对抗。

我们使用了团队基于Pogamut 2（Burkert等人，2007年）开发的工具。这些工具可在 svn://artemis.ms.mff.cuni.cz/pogamut/branches/fabienlatence 获取。实验为14位受试者每人提供了2个签名，每个签名包含20个成分。图2展示了部分结果样本。

结果分析

为了可视化数据，我们选择对签名进行主成分分析（PCA）。图3和图4使用两个主成分来展示受试者。

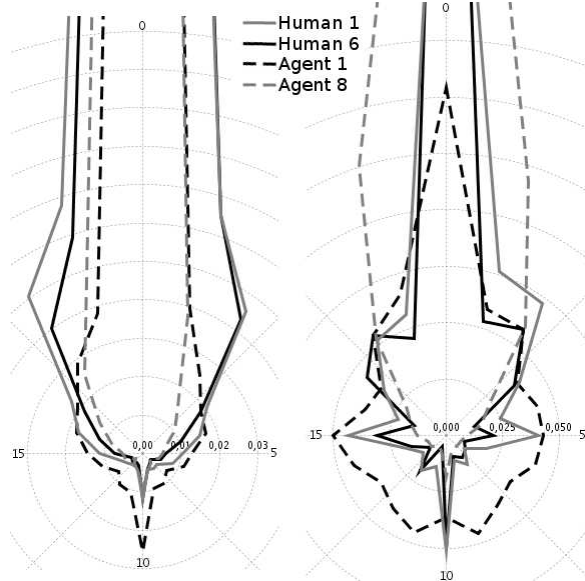


图2：签名样本。左侧为“速度变化角度”签名，右侧为“相对于方向的速度”签名。为了可读性，图形已对 0^{th} 分量进行截断显示。

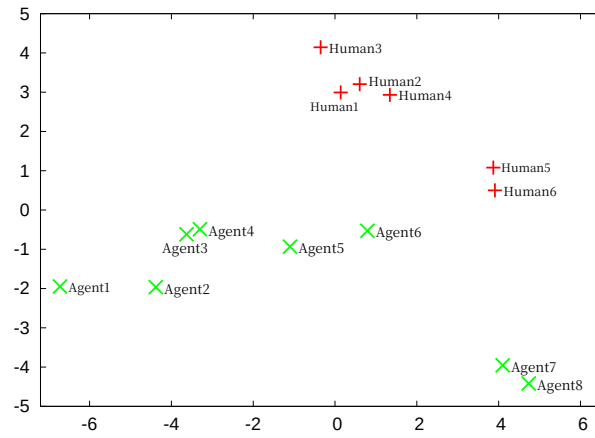


图3：使用两个主成分表示的受试者，这两个主成分解释了“速度变化角度”签名中 90.37% 的方差。

第一张图（图3）较好地呈现了数据，约 90% 的方差得到了体现。第一主成分轴与剧烈角度变化呈负相关，第二主成分轴与小角度变化呈正相关。第二张图（图4）精度稍低，仅体现了 78% 的方差；不过，它足以提取一些有趣的数据特征。第一主成分轴总体上与倒走以及向后+侧向移动呈负相关。第二主成分轴与向前+略带侧向的移动呈正相关，与向前+侧向移动呈负相关。

previous signature is used for angles. This vector is then normalised.

where t is the current time step and $t + 1$ is the following one, time steps occurring every 125 ms. $\vec{V}$ is the velocity vector without its z component, $\vec{D}$ is a vector pointing toward the character’s aiming point without its z component (figure 1). The z component was ignored for the signatures because it simplifies the signatures without losing too much information.

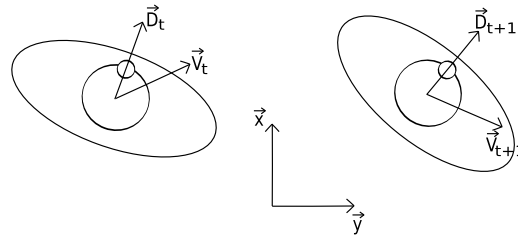


Figure 1: Scheme of a virtual character viewed from top view at two following time steps, t and $t + 1$. $\vec{D}$ is the direction vector, pointing toward the character’s aiming point and $\vec{V}$ is the velocity vector.

The experiment is composed of two steps:

1. 6 low/medium level human played a *deathmatch* game during 20 minutes during which signatures were computed. We chose a duration of 20 minutes because signatures tend to stabilise after 15 to 20 minutes. A *deathmatch* follows the basic rules and the objective is to “kill” as many other players as possible. Each human plays using a keyboard and a mouse to control the virtual character and a classic desktop computer screen to see the virtual environment.
2. Signatures are computed for 8 agents having 8 different skill levels. Those agents fight against each other in 6-agent matches in the same topography, following the same rules and during the exact same time.

We used tools developed by our team and based on Pogamut 2 (Burkert et al. 2007). They are available at svn://artemis.ms.mff.cuni.cz/pogamut/branches/fabienlatence. The experiment gave us 2 signatures for each of the 14 subjects, each signature having 20 components. A sample of those results is given in figure 2.

RESULTS’ ANALYSIS

In order to visualise the data, we chose to do a Principal Components Analysis (PCA) on the signatures. Subjects are represented using the two principal components in figures 3 and 4.

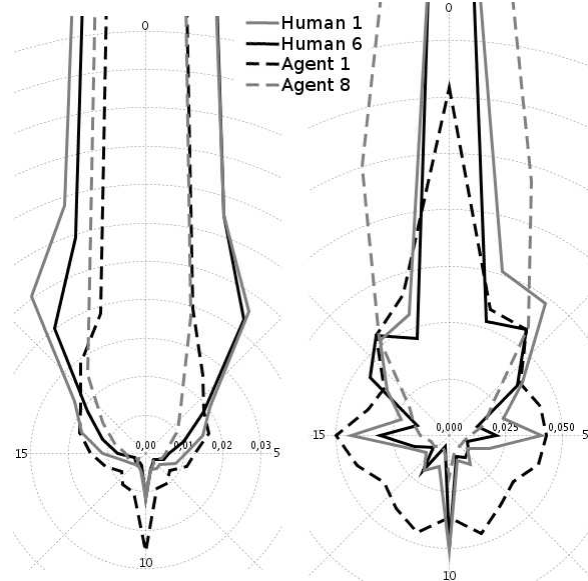


Figure 2: Sample of the signatures. On the left “velocity change angle” signatures and on the right “velocity relative to the direction” signatures. The graphs are truncated for the 0^{th} component for the sake of readability.

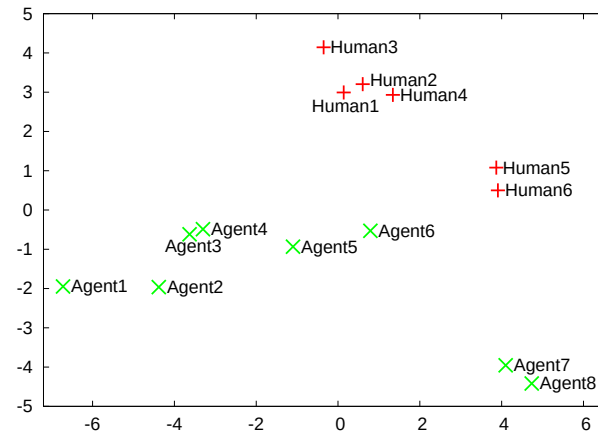


Figure 3: Subjects represented using the two principal components which explain 90.37% of the variance for the “velocity change angle” signatures.

The first chart (figure 3) represents very well the data with about 90% of the variance represented. The first PCA axis is negatively correlated with brutal angle changes and the second PCA axis is positively correlated with small angle changes. The second chart (figure 4) is less accurate with only 78% of the variance represented; however, it does suffice to extract some interesting data’s features. The first PCA axis is globally negatively correlated with walking backwards and backwards+sideways. The second PCA axis is positively correlated with walking forward+slightly sideways and negatively with forward+sideways.

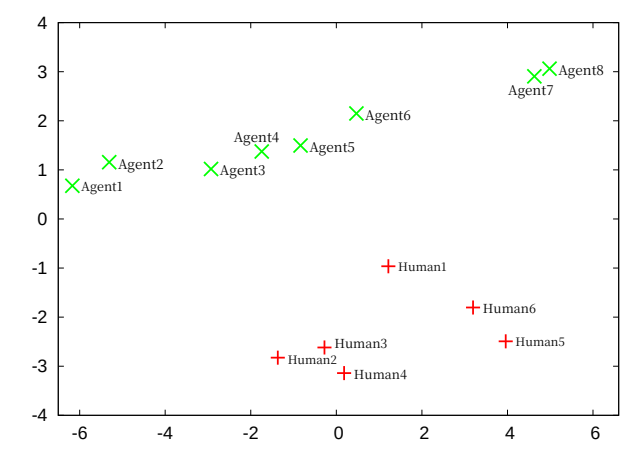


图4：使用两个主成分表示的受试者，这两个主成分解释了“相对于方向的速度”签名中78.18%的方差。

两张图揭示的最重要信息是，人类与智能体特征存在差异：在两张图中均可画出一条线将人类与智能体区分开来。为研究效率的影响，根据得分（“击杀数”）对智能体和人类进行排序。由于人类未与智能体对战，这些排名仅适用于各自群体内部：Human1 效率高于 Human2，Human2 效率高于 Human3，以此类推。Agent1 效率高于 Agent2，以此类推，但我们无法确定 Agent1 是否比 Human1 更高效。影响智能体效率的技能参数似乎产生了相当人为化的效果：在两张图中，智能体均沿第一主成分轴有序排列。

尽管这些结果很有前景，但主成分分析在我们的案例中存在一个缺陷：在签名中，某个分量 i 在角度上更接近 $i + 1^{th}$ 和 $i - 1^{th}$ ，而非 $i + 2^{th}$ 和 $i - 2^{th}$ 分量，而这一点并未被主成分分析所考虑。为了在考虑这一特性的情况下计算两个签名之间的距离，我们选择使用地球移动距离（EMD）（Rubner 等人，2000年）。打个比方，两个向量 V_1 和 V_2 之间的距离等于将“土堆” V_1 搬运到“土堆” V_2 所需的最小工作量。该度量的意义在图5中加以说明。

利用EMD，我们可以通过计算每个签名之间的距离来构建一个相异性矩阵。为了可视化这些相异性，我们采用了一种名为多维尺度分析（MDS）的方法，该方法将每个向量表示为平面中的一个点。MDS的目标是将给定的向量间相异性矩阵近似为表示这些向量的点之间的欧几里得距离矩阵（ L_2 ）。图6和图7展示了将MDS方法应用于由EMD

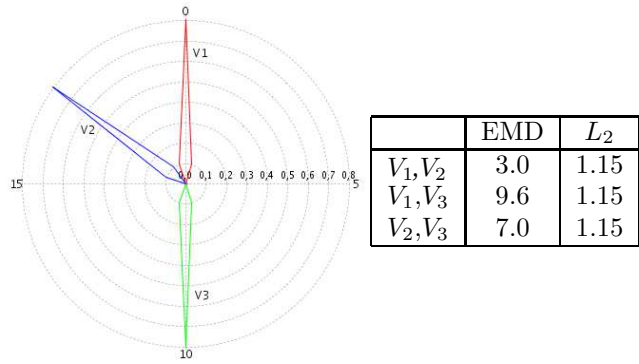


图5：示例展示了在我们的案例中，EMD相较于欧几里得距离（ L_2 ）的优势。由于归一化向量没有任何共同分量， L_2 距离彼此相等。然而，显然 V_1 和 V_2 是最相似的向量，这一点被EMD检测到。

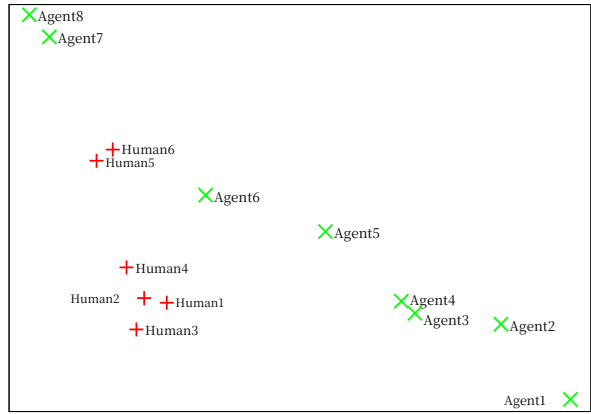


图 6：对“速度变化角度”签名计算出的EMD矩阵应用多维缩放的结果。该表示很好地体现了原始距离，因为压力略小于0.003。

计算得到的签名相异性矩阵的结果。请注意，这里不需要比例尺，因为重要的是点之间的相对距离，而非实际距离。

结果与主成分分析得到的略有不同：人类的签名与智能体特征混合得更紧密。在图7中，我们可以看到人类和智能体有着相当相似的行为，尽管两者之间仍存在一些差异。我们还可以注意到，在图6和图7中，所有智能体都位于一条直线上，这进一步强化了技能参数具有相当线性效应的观点。这种效应在图6的情况下似乎并不相关，因为随着人类效率的提升，他们往往会远离智能体特征。

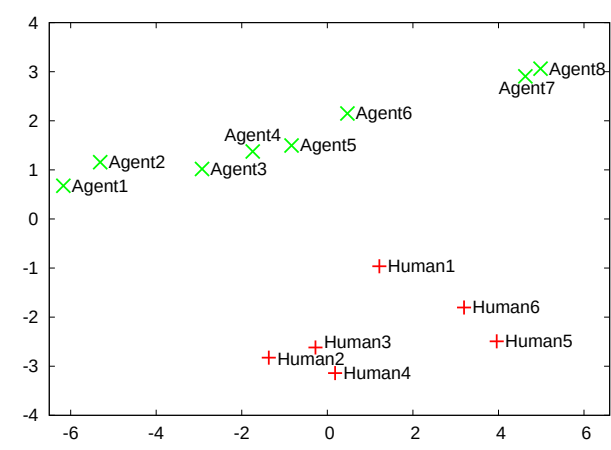


Figure 4: Subjects represented using the two principal components which explain 78.18% of the variance for the “velocity relative to the direction” signatures.

The most important information that both charts reveal is that there is a difference between humans’ and agents’ behaviours: it is possible to draw a line which separate humans from agents on the two charts. To study the influence of efficiency, agents and humans are ranked depending on their score (number of “kills”). As humans did not play against agents, these ranks apply only within those two groups: Human1 is more efficient than Human2, which is more efficient than Human3, etc. Agent1 is more efficient than Agent2, etc. but we do not know if Agent1 is more efficient than Human1. It seems that the *skill* parameter which influence the agents’ efficiency have a quite artificial effect: agents are ordered on the first principal component on both charts.

Even if those results are promising, PCA suffers from a flaw in our case: in the signatures, a component i is much closer in term of angle to the $i + 1^{th}$ and $i - 1^{th}$ than to the $i + 2^{th}$ and $i - 2^{th}$ component, which is not taken into account by the PCA. To calculate the distance between two signatures considering this particularity, we chose to use the Earth Mover’s Distance (EMD) (Rubner et al. 2000). Figuratively speaking, the distance between two vectors V_1 and V_2 is equal to the minimum effort made to carry earth from the “relief” V_1 to the “relief” V_2 . The interest of this metric is explained in the figure 5.

With the EMD, we can build a matrix of dissimilarities by computing the distance between each signature. To visualise these dissimilarities, we used a method called MultiDimentionnal Scaling (MDS) which consists in representing each vector by a point in a plan. The goal of MDS is to approximate a given dissimilarity matrix between vectors by a matrix of Euclidean distances (L_2) between the points representing the vectors. The result of the MDS method applied to dissimilarities ma-

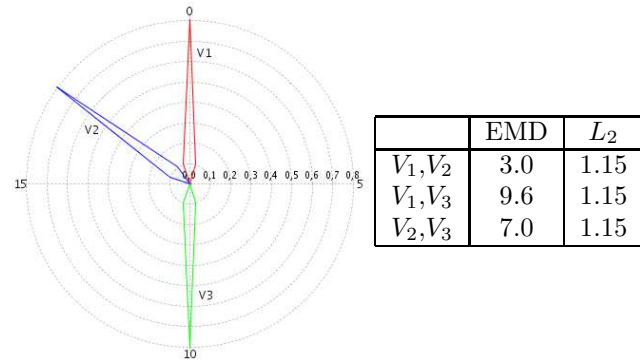


Figure 5: Example showing the interest of the EMD compared to Euclidean distance (L_2) in our case. The normalised vectors not having any common component, the L_2 distance is then equal between each other. However, it is obvious that V_1 and V_2 are the most similar vectors which is detected by the EMD.

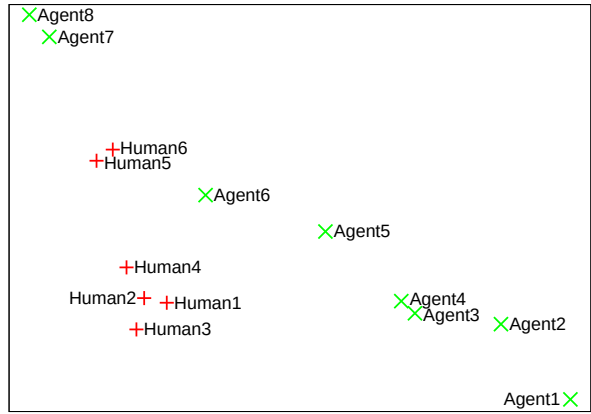


Figure 6: Result of a MDS applied to EMD matrix computed on “velocity change angle” signatures. This representation represents well the original distances as the stress is a bit less than 0.003.

trices computed with EMD on our signatures are given in figures 6 and 7. Note that there is no need for a scale because it is the relative distance between points that is important, not the real distance.

The results are a bit different from what we get with the PCA analysis: humans’ signatures are much more mixed with the agents’ than with the PCA. In the figure 7 we can see that humans and agents have quite similar behaviours although there are still some differences between the two. We can also note that all agents are on a line in figures 6 and 7 which reinforce the idea that the *skill* parameter have a quite linear effect. It seems that this effect is not pertinent in the case of the figure 6 because as the humans’ efficiency rise they tend to move away from the agents’.

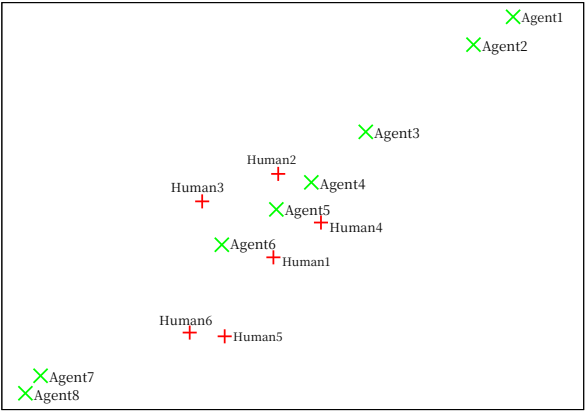


图7：对基于“相对于方向的速度”签名计算的EMD矩阵应用多维缩放的结果。该表示很好地反映了原始距离，因为压力值略高于0.003。

主成分分析与多维缩放分析似乎得出了截然不同的结果。多维缩放基于EMD，主要衡量全局签名形状之间的差异。从这个角度来看，智能体与人类相当接近，因此我们可以认为它们可能是可信的。然而，主成分分析更为严格，使我们能够清楚地区分人类和人工行为。这种差异可能源于人类使用键盘控制虚拟角色时存在的局限性。被评估的智能体并未复制这些局限性，因此它们可能表现出全局上类似人类的行为，但如果仔细观察，仍能被识别出来。需要注意的是，我们的两种签名旨在研究速度向量，因此所有结论仅适用于智能体的移动方式。为了研究行为中更多方面的特征，我们本应设计更多的签名。

结论

所提出的方法前景广阔，因为它有助于评估行为的可信度。其主要优势在于能够评估大量智能体，从而更精细地改进模型参数。这一优势源于签名原理——即表征行为各个方面的向量。我们在测试实验中发现，即使是简单的签名也能产生有趣的结果。

然而，仍有一些问题有待解答：人类在多大程度上能察觉行为中的变化？签名是否具有与人类相同的灵敏度？应将当前方法与经典方法进行对比研究，以评估签名的有效性。即便此类研究耗时且复杂，它们也将产生有用且可复用的签名。

下一步是将该方法应用于行为建模项目。其目标是根据基于本工作的评估方法来优化模型。最终智能体将通过经典方式进行评估，由受试者在环境中对智能体进行评判。

然而，我们需要改进签名的类型，以使评估更加精确。目前展示的签名是全局签名，它们在每个时间步长进行计算。而上下文签名则仅在满足特定感知和/或动作条件时才进行计算，这将提供关于行为更有意义的信息。另一种新的签名类型也可能非常有用，即时间签名，用于测量事件之间的时间间隔。它可以用来测量反应时间，这是可信度的一个重要因素。

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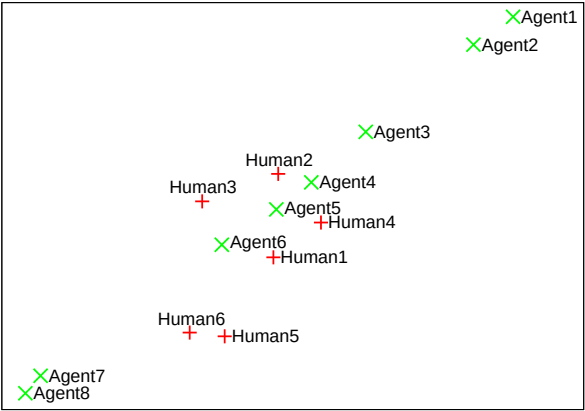


Figure 7: Result of a MDS applied to EMD matrix computed on “velocity relative to the direction” signatures. This representation represents well the original distances as the stress is a bit more than 0.003.

The PCA and the MDS analysis seems to give quite different results. The MDS is based on the EMD which measure the difference between global signatures’ shapes. From this point of view, agents are quite close to humans so we can think that they might be believable. However, the PCA analysis, being more strict, allows us to clearly distinguish human and artificial behaviours. This difference can come from the fact that humans have limitations to control their virtual character because they use keyboards. Evaluated agents do not copy those limitations, as a result, they may have a global human-like behaviour but they can be recognised if we look closely. Note that our two signatures are designed to study the velocity vector so all our conclusion applies only to the way agents move. We should have designed a lot more signatures in order to study much more aspects of the behaviours.

CONCLUSION

The proposed method seems promising as it could help in assessing the believability of a behaviour. Its main advantage is that it can evaluate a large number of agents, allowing finer improvement of the models’ parameters. This advantage is due to the principle of signatures, vectors which characterise behaviours’ aspects. We found out in our test experiment that even simple signatures can give interesting results.

However, there is still some questions that should be answered: to what extent do humans notice variations in behaviours? and do signatures have the same sensitivity as humans? Studies confronting the current method to classic ones should be done to evaluate signatures. Even if those studies are time-consuming and complex they will result in useful and reusable signatures.

The next step is to use this method in a behaviour modelling project. The goal is to optimise the models according to an evaluation method based on our present work. The final agents will then be evaluated in a classic way, with subjects judging agents in their environment.

We will, however, need to improve the type of signatures for the evaluation to be more precise. Presented signatures are *global signatures*, they are computed at each time step. *Contextual signatures* are computed only when certain perceptions and/or actions based conditions are met. They will provide more meaningful information about behaviours. Another new type of signature can be useful, *temporal signatures* which will measure reaction time between events. It can be used to measure reaction time which is an important factor of believability.

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